



New spectral lower bounds on the bisection width of graphs[☆]

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Abstract

The communication overhead is a major bottleneck for the execution of a process graph on a parallel computer system. In the case of two processors, the minimization of the communication can be modeled using the graph bisection problem. The spectral lower bound of $\lambda_2|V|/4$ for the bisection width of a graph is widely known. The bisection width is equal to $\lambda_2|V|/4$ iff all vertices are incident to $\lambda_2/2$ cut edges in every optimal bisection.

We present a new method of obtaining tighter lower bounds on the bisection width. This method makes use of the level structure defined by the bisection. We define some global expansion properties and we show that the spectral lower bound increases with this global expansion. Under certain conditions we obtain a lower bound depending on $\lambda_2^\beta|V|$ with $\frac{1}{2} \leq \beta < 1$. We also present examples of graphs for which our new bounds are tight up to a constant factor. As a by-product, we derive new lower bounds for the bisection widths of 3- and 4-regular Ramanujan graphs.

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1. Introduction

The relation between the second smallest eigenvalue of the Laplacian and the bisection width of a graph has already been discussed in literature. A bisection of a graph is an equal distribution of the vertices of a graph into two parts. The cut size is the number of crossing edges between the two parts. The bisection width is the minimal cut size of all bisections. Its calculation for arbitrary graphs is NP-complete [10], even when it is restricted to the class of d -regular graphs [4]. The bisection of a graph and, more generally, the partitioning of a graph into several balanced parts have a wide range of applications, especially in the field of parallel computation.

Many different methods of calculating partitions of a graph have been proposed in the past by scientists from different fields such as mathematics, computer science and engineering. There are also many software libraries available which include the most efficient graph partitioning approaches. Most heuristics have been widely used in many applications and the partition qualities are satisfactory.

This is not the case for lower bounds on the bisection width and only a few approaches to this problem have so far been developed. Leighton [13] proposes a lower bound of the bisection width by calculating a routing scheme between all pairs of vertices such that the congestion is minimized. This lower bound is strict for some graphs such as complete graphs, hypercubes, grids and tori. However, for irregularly structured graphs, as they appear in most applications, the problem of calculating a routing scheme with a low congestion remains.

Lower bounds on the bisection width can also be derived from algebraic graph theory by relating the bisection problem to an eigenvalue problem. More precisely, there is a relationship between the bisection width of a graph $G = (V, E)$ and the second smallest eigenvalue of its so called Laplacian $L(G) = \{l_{v,w}\}$ which is defined as follows:

$$l_{v,w} = \begin{cases} \deg(v) & \text{if } v = w, \\ -1 & \text{if } v \neq w \text{ and } \{v, w\} \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The widely known lower bound on the bisection width $bw(G)$ of a graph $G = (V, E)$, where $n = |V|$ is even and with the second smallest eigenvalue λ_2 of $L(G)$ is

$$bw(G) \geq \frac{\lambda_2 n}{4}. \quad (1)$$

In Theorem 1, we show that the equality holds for (1) iff all vertices are incident to exactly $\frac{\lambda_2}{2}$ cut edges. This is the case for some graphs, e.g. complete graphs, complete bipartite graphs, hypercubes and the Petersen graph. Nevertheless, there is a large gap between this lower bound and the bisection width for most graphs. To give an example, crash or flow simulations can be solved using the finite element method. The mesh used to discretize the object often has a structure which is similar to that of a grid. The second smallest eigenvalue of a $\sqrt{n} \times \sqrt{n}$ grid is $\lambda_2 = 2 - 2 \cos(\pi/\sqrt{n}) \approx \pi^2/n$. Obviously, the bisection width is $\sqrt{n}$. However, the lower bound of Eq. (1) only results in a value of $\frac{\pi^2}{4}$. Here, a relation between bw and $\sqrt{\lambda_2}$ would give a tight bound.

Such a quadratical gap also appears in the relation between the isoperimetric number and λ_2 [5,7,16]. The isoperimetric number of a graph $G=(V,E)$ is defined as $i(G)=\min_{S\subset V} \frac{|E(S,\bar{S})|}{\min\{|S|,|\bar{S}|\}}$, where $E(S,\bar{S})$ is the set of edges which connects vertices of S with vertices of $\bar{S}$. In graphs with a large isoperimetric number, information can flow more easily from one part of the graph to the other. In graphs with a large bisection width the information can flow easily between two parts of the same size, however some bottlenecks can still occur in one of these parts. Since the bisection width of a graph only involves sets of size $|V|/2$, it holds that $i(G)\leq 2bw(G)/n$. In [7,16] it is shown that

$$\frac{i^2(G)}{2d_{\max}(G)} \leq \lambda_2 \leq 2i(G)$$

with maximal degree $d_{\max}(G)$. Additionally, $\lambda_2 = O(\frac{1}{n})$ for bounded-degree planar graphs and two-dimensional meshes and $\lambda_2 = O(\frac{1}{n^{2/d}})$ for well-shaped d -dimensional meshes [21].

If the lower bound of Eq. (1) is not strict, either all vertices are still incident to cut edges but not to exactly $\frac{\lambda_2}{2}$ of them, or there are vertices which are not incident to any cut edge. We will present a new method of determining an upper bound for λ_2 and this method makes use of the level structure of a bisection in which each level consists of all the vertices which have the same distance to the cut. This will lead to a generalized lower bound of Eq. (1), depending on the growth of the sizes of these levels on each side of the cut. Let $g:\mathbb{N}\rightarrow\mathbb{N}$ be a function. We will introduce the class of level-structured graphs $LS(g,\sigma)$, which have a bisection with a cut size of σ and a level structure, such that there are no more than $\sigma g(i)$ edges between vertices of distance i and vertices of distance $i+1$ to the cut. We show that if the sum $1+2\sum_{i=2}^{\infty} \frac{1}{g(i-1)}$ is a constant A , then for $\frac{n}{\sigma}\rightarrow\infty$ holds that

$$\sigma \geq A \frac{\lambda_2 n}{4} (1 + o(1)). \quad (2)$$

In the case of $g(i)=(i+1)^\alpha$, $0\leq\alpha<1$, we show that for $\frac{n}{\sigma}\rightarrow\infty$ holds that

$$\sigma \geq \frac{1+\alpha}{2((1-\alpha)(3-\alpha))^{\frac{\alpha+1}{2}}} \lambda_2^{\frac{\alpha+1}{2}} n(1+o(1)). \quad (3)$$

Thus, we established a relation between σ and $\lambda_2^\beta n$ in the range $\frac{1}{2}\leq\beta<1$. For Eq. (2) we show that there are graphs from $LS(g,\sigma)$ for which the bound is tight and for (3), we show that there are graphs for which the bound is tight up to a small constant factor. By using similar methods, we were able to generalize the bounds for the k -section width of graphs for arbitrary k [8]. The proofs used here might be of special interest to some readers since we developed new techniques for computing asymptotic exact the second smallest eigenvalue of the Laplacian for certain weighted paths.

Eqs. (2) and (3) may also be used the other way around to derive an upper bound of λ_2 from any bisection with a cut size σ and growing function g .

We achieve an improved relation between the bisection width and λ_2 if the growth of the level sizes which are defined by an optimal bisection is bounded by some function $g(i)$. In general, the level structure of an optimal bisection is not known, but there is one notable exception. If G is a graph of maximum degree d , then $G \in LS(g, bw(G))$ where $g(i) = (d-1)^i$. Thus, for $\frac{n}{bw} \rightarrow \infty$, Eq. (2) results in a lower bound of

$$bw \geq \frac{d}{d-2} \frac{\lambda_2 n}{4} (1 + o(1)).$$

We expect that further classes of graphs exist for which Eq. (2) or (3) can be directly used to derive a lower bound on the bisection width.

As a by-product, we get better lower bounds on the bisection width of d -regular Ramanujan graphs, where $\lambda_2 \geq d - 2\sqrt{d-1}$ and, thus, for a fixed d it holds that $bw = \Theta(n)$. There are several methods of explicitly constructing Ramanujan graphs (see e.g. [6,14,15,17]). Eq. (1) leads to lower bounds of $0.043n$ and $0.134n$ for the bisection widths of 3- and 4-regular Ramanujan graphs, respectively. With the use of the level structure of a bisection, we can prove that any 3-regular Ramanujan graph has a bisection width of at least $0.082n$ and any 4-regular Ramanujan graph a bisection width of at least $0.176n$. These are the highest lower bounds for explicitly constructible 3- and 4-regular graphs. However, 3- and 4-regular graphs with a higher lower bound on the bisection width also exist. Kostochka and Melnikov [12] showed that the bisection width is at least $\frac{1}{4.95} n \approx 0.202n$ for almost all 3-regular random graphs and Bollobas [3] showed that the bisection width is at least $\frac{11}{50} n = 0.22n$ for almost all 4-regular random graphs.

Section 2 presents the basic relation between λ_2 and the bisection width of the graph. In Section 3, we show a new relation between λ_2 and the cut size σ of a bisection. In Section 4, we demonstrate that there are graphs for which the new bounds are tight up to a constant.

Some of the results contained in this paper have already been presented at the Workshop on Graph Theoretic Concepts in Computer Science (WG'00) (see [2]).

2. Definitions and background

A bisection π of an undirected graph $G = (V, E)$ where $|V| = n$ is a partition $V = V_0 \cup V_1$ such that $|V_0|, |V_1| \in \{\lfloor n/2 \rfloor, \lceil n/2 \rceil\}$. In the rest of the paper we assume that n is even. For a bisection π , the number of edges between V_0 and V_1 is called the *cut size* of π . The minimum cut size of a bisection is called the *bisection width* $bw(G)$ of G , which is formally defined as $bw(G) := \min\{|\{v, w\} \in E; \pi(v) \neq \pi(w)\}|; \pi \text{ is bisection of } G\}$.

It is known that all the eigenvalues of the Laplacian $L(G)$ of a graph $G = (V, E)$ are nonnegative. We denote them as $\lambda_1, \dots, \lambda_n$ with $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ which have pairwise perpendicular eigenvectors $y_1, y_2, \dots, y_n$, where $y_1 = (1, 1, \dots, 1)$ is an eigen vector to the eigenvalue 0. The multiplicity of this eigenvalue 0 is equal to the number of connected components in the graph, i.e., $\lambda_i = 0$ for any $i \in \{1, \dots, c(G)\}$, where $c(G)$ represents the number of connected components in G (see e.g. [20]). However, we

only consider connected graphs, so $\lambda_2 > 0$. Let $x = (x_1, x_2, \dots, x_n)$ be a non-zero vector. It follows from the Courant–Fisher principle that

$$\lambda_2 = \min_{x \perp 1} \left\{ \frac{x^T L x}{\|x\|^2} \right\} = \min_{x \perp 1} \left\{ \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{\sum_{v \in V} x_v^2} \right\}. \quad (4)$$

Furthermore, the minimum in (4) is attained iff x is an eigenvector to λ_2 . Using the Lagrange identity, (4) can be rewritten (cf. [9]) as

$$\lambda_2 = \min_{x \neq \text{const}} \left\{ \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{\sum_{\{u,v\} \in V^2} (x_u - x_v)^2} n \right\}. \quad (5)$$

Here the minimum runs over all vectors which are not collinear to $(1, 1, \dots, 1)$. Now a simple lower bound on $bw(G)$ can be derived by applying (5) to the x -tuple which is defined as

$$x_v = \begin{cases} a & \text{if } v \in V_0, \\ b & \text{if } v \in V_1. \end{cases} \quad \text{with some } a \neq b. \quad (6)$$

This leads to $\lambda_2 \leq n \frac{bw(a-b)^2}{(n/2)(n/2)(a-b)^2} = \frac{4bw}{n}$, i.e. to the well-known lower bound of $bw \geq \frac{\lambda_2 n}{4}$, which is strict for some classes of graphs. In the following theorem we thoroughly specify the situation when this bound is attainable.

Theorem 1. *Let $G = (V, E)$ be a graph and λ_2 be the second smallest eigenvalue of $L(G)$. Then the following statements are equivalent:*

- (a) $bw(G) = \frac{\lambda_2 n}{4}$;
- (b) *there is an eigenvector corresponding to λ_2 which has only -1 and $+1$ entries;*
- (c) *in any optimal bisection $V = V_0 \cup V_1$ any vertex is incident to exactly $\frac{\lambda_2}{2}$ cut edges.*

Proof. We prove that statement (a) implies (b), (b) implies (c), and (c) implies (a).

Assume $bw(G) = \frac{\lambda_2 n}{4}$ and let $V = V_0 \cup V_1$ be an optimal bisection. Consider the vector $(x_1, \dots, x_n)$ where $x_u = 1$ for $u \in V_0$, and $x_u = -1$ for $u \in V_1$. Now (5) implies

$$\lambda_2 \leq \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{\sum_{\{u,v\} \in V^2} (x_u - x_v)^2} n = \frac{4bw(G)}{n} = \lambda_2.$$

Therefore, x is an eigenvector which corresponds to λ_2 .

In order to prove the second implication, we assume there is an eigenvector x which corresponds to λ_2 specified above. Let $u \in V_0$ (a similar argument works well for $u \in V_1$), and let a_u (resp. b_u) denote the number of vertices of V_0 (resp. V_1) incident to u . Then the u th entry of $L(G)x$ equals $\deg(u) - a_u + b_u$. Since $L(G)x = \lambda_2 x$ and the u th entry of x is 1, then $\deg(u) - a_u + b_u = \lambda_2$. This equality along with $a_u + b_u = \deg(u)$ implies that $b_u = \frac{\lambda_2}{2}$.

Finally, let $V = V_0 \cup V_1$ be an optimal bisection, and assume that any vertex is incident to exactly $\frac{\lambda_2}{2}$ cut edges. Since $|V_0| = |V_1| = n/2$, then the size of the cut equals

$\frac{n}{2} \frac{\lambda_2}{2}$. On the other hand the size of the cut equals $bw(G)$ because the bisection is optimal. $\square$

Therefore, there are plenty of graphs for which $bw(G) = \frac{\lambda_2 n}{4}$ holds. Among examples of such graphs are the complete graphs where $\lambda_2 = n$ and with a bisection width $bw = n^2/4$, the complete bipartite graphs where $\lambda_2 = \frac{n}{2}$ and $bw = \frac{n^2}{8}$, the hypercubes where $\lambda_2 = 2$ and $bw = n/2$ or the Petersen graph ($n = 10$), where $\lambda_2 = 2$ and $bw = 5$.

3. New relations between λ_2 and the bisection width

As we have already shown in Theorem 1, the lower bound (1) is only tight if any vertex of G is incident to a cut edge. Now, we consider the case where this condition is not satisfied, and we show that this lower bound can be significantly improved for such graphs. Therefore, we consider the level structure of a bisection. Each level consists of all the vertices which have the same distance to the cut.

Definition 1 (Level structure). Let π be a bisection of a graph $G = (V, E)$, where $V = V_0 \cup V_1$ and with a cut size σ . The subsets V_0^i of V_0 are defined as follows:

Let V_0^1 be the set of all vertices in V_0 that are incident to a cut edge. Let V_0^i be the sets of all vertices in V_0 with distance $i - 1$ to a vertex of V_0^1 . The respective sets on side V_1 are denoted as V_1^i . Furthermore, the edge sets which connect vertices between V_ε^i and V_ε^{i+1} are denoted as E_ε^i , where $i \geq 1$ and $\varepsilon \in \{0, 1\}$.

Let $g: \mathbb{N} \rightarrow \mathbb{N}$ be a function. We denote the class of graphs which have a bisection π with a cut size of σ and a level structure of the kind that $|E_0^i| \leq \sigma g(i)$ and $|E_1^i| \leq \sigma g(i)$ for all $i \geq 1$ as $LS(g, \sigma)$.

To a certain extent, the level structure can be regarded as a double cone. The function g represents its width. In the case of grids and paths it holds that $g(i) = 1$ for any i . For example, the levels of an 8×8 square grid and a median cut is shown in Fig. 1(left). For graphs with a cone of larger width (as in Fig. 1(right)), information can flow more easily to the distant vertices. Therefore, we can view this as a global expansion property.

In the rest of this section, we show that for a fixed bisection width the spectral lower bound of the second smallest eigenvalue increases with the width of the cone.

In Lemma 1 we bound λ_2 from above using an expression which only depends on the grow function $g(i)$. Indeed, the proof of the lemma shows that the worst case scenario occurs if $|V_i| = \sigma g(i - 1)$ holds for any level i . This result will be used in Theorem 2 below.

Lemma 1. Let $G \in LS(g, \sigma)$ and let $l \in \mathbb{N}$ be a number such that $n \geq 2\sigma \sum_{i=1}^{l-1} g(i - 1)$. Then

$$\lambda_2 \leq \min_{1=a_1 \leq a_2 \leq \dots \leq a_l} \frac{2\sigma a_1^2 + \sigma \sum_{i=1}^{l-1} g(i)(a_i - a_{i+1})^2}{\sigma \sum_{i=1}^{l-1} a_i^2 g(i - 1) + a_l^2 \left(\frac{n}{2} - \sigma \sum_{i=1}^{l-1} g(i - 1) \right)}.$$

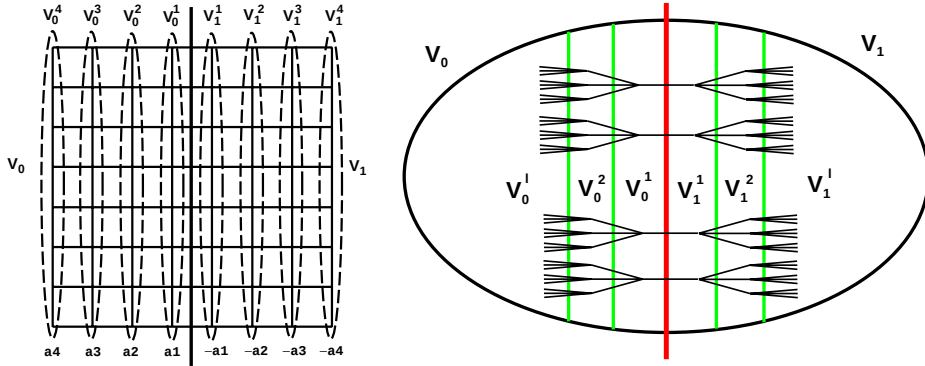


Fig. 1. Left: levels of an 8×8 square grid and a median cut. Right: The number of edges connecting consecutive levels may increase with the function $g(i)$. The assignment of the a_i -values is defined in Eq. (7) of Lemma 1.

Proof. In the proof we use Eqs. (4) and (5). We need a non-zero vector x with $x \perp 1$ for Eq. (4), whereas we only need a non-constant x for (5).

First, we use Eq. (5) in such a way that all vertices of the same level have identical values in the entries of the vector. We choose a vector x with entries x_v for $v \in V$ which are defined as

$$x_v = \begin{cases} a_i & \text{if } v \in V_0^i \text{ with } i < l, \\ a_l & \text{if } v \in V_0^i \text{ with } i \geq l, \\ -a_i & \text{if } v \in V_1^i \text{ with } i < l, \\ -a_l & \text{if } v \in V_1^i \text{ with } i \geq l, \end{cases} \quad (7)$$

where $1 = a_1 \leq a_2 \leq \dots \leq a_l$. The upper bound of Eq. (5) now only depends on the a_i . Let $A(x) := \sum_{\{u,v\} \in E} (x_u - x_v)^2$ be the numerator of (4) and (5). Then, it holds that

$$\begin{aligned} A(x) &= 4\sigma a_1^2 + \sum_{i=1}^{l-1} |E_0^i| (a_i - a_{i+1})^2 + \sum_{i=1}^{l-1} |E_1^i| (a_i - a_{i+1})^2 \\ &\leq 4\sigma a_1^2 + 2\sigma \sum_{i=1}^{l-1} g(i) (a_i - a_{i+1})^2. \end{aligned} \quad (8)$$

Now we estimate the denominator $B(x) := \sum_{\{u,v\} \in V^2} (x_u - x_v)^2$ of (5). It holds that $|V_0^j| \leq |E_0^{j-1}| \leq \sigma g(j-1)$. We assume now that $|V_0^j| < \sigma g(j-1)$ for some j , $1 \leq j < l$. In the sequel, we merge all the levels V_e^i for any $i \geq l$ to one level and we denote this as V_e^l . The corresponding vector for the level structure which is obtained by moving a vertex from the merged level V_0^l to V_0^j is $\tilde{x}$. We then have

$$\begin{aligned} B(\tilde{x}) - B(x) &= - \sum_{i=1}^{l-1} (a_i - a_l)^2 |V_0^i| + \sum_{i=1}^l (a_i - a_j)^2 |V_0^i| - (a_l - a_j)^2 \\ &\quad + \sum_{i=1}^l ((a_i + a_j)^2 - (a_i + a_l)^2) |V_1^i| \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^{l-1} ((a_i - a_j)^2 - (a_i - a_l)^2) |V_0^i| + (a_l - a_j)^2 |V_0^l| - (a_l - a_j)^2 \\
&\quad + \sum_{i=1}^l ((a_i + a_j)^2 - (a_i + a_l)^2) |V_1^i| \\
&= \sum_{i=1}^{l-1} ((a_i - a_j)^2 - (a_i - a_l)^2) |V_0^i| + (a_l - a_j)^2 \left(\sum_{i=1}^l |V_0^i| - \sum_{i=1}^{l-1} |V_0^i| \right) \\
&\quad - (a_l - a_j)^2 + \sum_{i=1}^l ((a_i + a_j)^2 - (a_i + a_l)^2) |V_1^i| \\
&= \sum_{i=1}^{l-1} ((a_i - a_j)^2 - (a_i - a_l)^2) |V_0^i| + (a_l - a_j)^2 \sum_{i=1}^l |V_1^i| \\
&\quad - (a_l - a_j)^2 \sum_{i=1}^{l-1} |V_0^i| - (a_l - a_j)^2 + \sum_{i=1}^l ((a_i + a_j)^2 - (a_i + a_l)^2) |V_1^i| \\
&= \sum_{i=1}^{l-1} ((a_i - a_j)^2 - (a_i - a_l)^2 - (a_l - a_j)^2) |V_0^i| - (a_l - a_j)^2 \\
&\quad + \sum_{i=1}^l ((a_i + a_j)^2 - (a_i + a_l)^2 + (a_l - a_j)^2) |V_1^i| \\
&= \sum_{i=1}^{l-1} 2(a_l - a_j)(a_i - a_l) |V_0^i| - (a_l - a_j)^2 \\
&\quad + \sum_{i=1}^l 2(a_i + a_j)(a_j - a_l) |V_1^i| \\
&\leq 0.
\end{aligned}$$

In order to establish the third and fourth equality, we use $|V_0^l| = \sum_{i=1}^l |V_0^i| - \sum_{i=1}^{l-1} |V_0^i|$
 $= \sum_{i=1}^l |V_1^i| - \sum_{i=1}^{l-1} |V_0^i|$. Here we only assume that $|V_0| = |V_1|$ (we do not necessary
need to have $|V_0^i| = |V_1^i|$). The inequality follows from $a_l \geq a_i$ for $0 \leq i \leq l$. Therefore,
 $B(x)$ does not increase because of this transformation and the minimum of $B(x)$ (for a
fixed V_1 part) is attained if $|V_0^i| = \sigma g(i-1)$, where $1 \leq i < l$. Using similar arguments,
by moving a vertex from $|V_1^l|$ to $|V_1^j|$ for some $j \in \{1, \dots, l-1\}$, we can show that $B(x)$
is minimized if $|V_1^i| = \sigma g(i-1)$ for $1 \leq i < l$. Therefore, $B(x)$ is minimized if $|V_0^i| = |V_1^i|$
for $1 \leq i < l$. Then, $x \perp 1$ and the denominators of (4) and (5) (when multiplied by $\frac{1}{n}$)
are equal. These arguments imply that

$$\sum_{v \in V} x_v^2 \geq 2\sigma \sum_{i=1}^{l-1} a_i^2 g(i-1) + 2a_l^2 \left(\frac{n}{2} - \sigma \sum_{i=1}^{l-1} g(i-1) \right). \quad (9)$$

The lemma follows on from this by substituting (8) and (9) into (4). $\square$

Lemma 1 shows that the level structure of the bisection gives an upper bound on λ_2 . We will use this result in the following theorem to derive new relations between

λ_2 and the cut size σ of a bisection. These new relations depend on the growth of the function g .

Theorem 2. Let $G \in LS(g, \sigma)$. Then a function $\gamma: \mathbb{R}^+ \rightarrow \mathbb{R}$ with $\gamma(x) \rightarrow 0$ for $x \rightarrow \infty$ exists such that

- if $A := 1 + 2 \sum_{i=2}^{\infty} \frac{1}{g(i-1)} < \infty$, then $\sigma \geq A \frac{\lambda_2 n}{4} (1 + \gamma(\frac{n}{\sigma}))$.
- if $g(i) = (i + 1)$, then $\sigma \geq \text{Lambert}W(\frac{4}{\lambda_2}) \frac{\lambda_2 n}{4} (1 + \gamma(\frac{n}{\sigma}))$, where $\text{Lambert}W(x)$ is the inverse function of xe^x .
- if $g(i) = (i + 1)^\alpha$ and $0 \leq \alpha < 1$, then $\sigma \geq \delta(\alpha) \lambda_2^{\frac{\alpha+1}{2}} n (1 + \gamma(\frac{n}{\sigma}))$ where $\delta(\alpha) = \frac{1+\alpha}{2((1-\alpha)(3-\alpha))^{\frac{\alpha+1}{2}}}$.

Proof. Let $l \in \mathbb{N}$ be defined by the inequality

$$2\sigma \sum_{i=1}^{l-1} g(i-1) < n \leq 2\sigma \sum_{i=1}^l g(i-1).$$

This, in particular, implies that $\frac{n}{\sigma} \rightarrow \infty$ as $l \rightarrow \infty$. We apply $a_i = 1 + 2 \sum_{j=2}^i \frac{1}{g(j-1)}$ to Lemma 1. Since

$$2a_1^2 + \sum_{i=1}^{l-1} g(i)(a_i - a_{i+1})^2 = 2 + 4 \sum_{i=1}^{l-1} \frac{1}{g(i)} = 2a_l$$

then

$$\lambda_2 \leq \frac{2\sigma a_l}{\sigma \sum_{i=1}^{l-1} a_i^2 g(i-1) + a_l^2 \left(\frac{n}{2} - \sigma \sum_{i=1}^{l-1} g(i-1) \right)} \leq \frac{2a_l}{\sum_{i=1}^{l-1} a_i^2 g(i-1)}. \quad (10)$$

In the sequel we use some functions γ_i with the same property as those described for γ

- $A := 1 + 2 \sum_{i=2}^{\infty} \frac{1}{g(i-1)} < \infty$: Set $r_i := A - a_i$, $1 \leq i \leq l$, where $r_i > 0$. The first inequality of (10) implies that

$$\begin{aligned} \lambda_2 &\leq \frac{2\sigma(A - r_l)}{\sigma \sum_{i=1}^{l-1} (A^2 - 2Ar_i + r_i^2)g(i-1) + (A^2 - 2Ar_l + r_l^2) \left(\frac{n}{2} - \sigma \sum_{i=1}^{l-1} g(i-1) \right)} \\ &= \frac{2\sigma(A - r_l)}{\sigma A^2 (1 + \gamma_1(\frac{n}{\sigma})) \sum_{i=1}^{l-1} g(i-1) + A^2 (1 + \gamma_2(\frac{n}{\sigma})) \left(\frac{n}{2} - \sigma \sum_{i=1}^{l-1} g(i-1) \right)} \\ &\leq \frac{4\sigma}{An(1 + \gamma(\frac{n}{\sigma}))}. \end{aligned}$$

- $g(i) = (i + 1)$: Note that $\ln(n + 1) = \int_1^{n+1} \frac{1}{x} dx \leq \sum_{i=1}^n \frac{1}{i} \leq 1 + \int_1^n \frac{1}{x} dx = 1 + \ln(n)$. Therefore, the second inequality in (10) where $g(i) = i + 1$ provides

$$\begin{aligned} \lambda_2 &\leq \frac{2a_l}{\sum_{i=1}^{l-1} i a_i^2} \leq \frac{4 \ln(l) + 2}{1 + \sum_{i=2}^{l-1} i(2 \ln(i + 1) - 1)^2} \\ &\leq \frac{4 \ln(l) + 2}{1 + \int_1^{l-1} (4x \ln^2(x + 1) - 4x \ln(x + 1) + x) dx} \leq \frac{2}{l^2 \ln(l)(1 + \gamma_1(\frac{n}{\sigma}))}. \end{aligned}$$

It follows from $n \leq 2\sigma \sum_{i=1}^l i = 2\sigma \frac{l^2}{2} (1 + \gamma_2(\frac{n}{\sigma}))$ that $l \geq \sqrt{\frac{n}{\sigma}(1 + \gamma_2(\frac{n}{\sigma}))}$. This leads to $\lambda_2 \leq \frac{2\sigma}{n \ln(\sqrt{n/\sigma})} (1 + \gamma_2(\frac{n}{\sigma}))$. When this equation is solved in σ we obtain the result of the theorem.

- $g(i) = (i + 1)^\alpha$ where $0 \leq \alpha < 1$: Since

$$\sum_{i=1}^l i^\beta = \frac{l^{\beta+1}}{\beta+1} (1 + o(1)) \text{ for } \beta > -1,$$

it holds that

$$\begin{aligned} \lambda_2 &\leq \frac{2a_l}{\sum_{i=1}^{l-1} i^\alpha a_i^2} \leq \frac{4 \frac{l^{1-\alpha}}{1-\alpha}}{\sum_{i=1}^{l-1} i^\alpha 4(\frac{i^{1-\alpha}}{1-\alpha})^2 (1 + \gamma_1(\frac{n}{\sigma}))} = \frac{(1 - \alpha) l^{1-\alpha}}{\sum_{i=1}^{l-1} i^{2-\alpha} (1 + \gamma_1(\frac{n}{\sigma}))} \\ &\leq \frac{(1 - \alpha)(3 - \alpha)}{l^2 (1 + \gamma_2(\frac{n}{\sigma}))}. \end{aligned}$$

From $n \leq 2\sigma \sum_{i=1}^l i^\alpha = 2\sigma \frac{l^{\alpha+1}}{\alpha+1} (1 + \gamma_3(\frac{n}{\sigma}))$ it follows that $l \geq (\frac{n(\alpha+1)}{2\sigma(1+\gamma_3(\frac{n}{\sigma}))})^{\frac{1}{\alpha+1}}$. This leads to

$$\begin{aligned} \lambda_2 &\leq \frac{(1 - \alpha)(3 - \alpha)}{(\frac{n(\alpha+1)}{2\sigma})^{\frac{2}{\alpha+1}} (1 + \gamma_4(\frac{n}{\sigma}))}, \text{ i.e.} \\ \sigma &\geq \frac{1 + \alpha}{2((1 - \alpha)(3 - \alpha))^{\frac{\alpha+1}{2}}} \lambda_2^{\frac{\alpha+1}{2}} n \left(1 + \gamma\left(\frac{n}{\sigma}\right)\right). \quad \square \end{aligned}$$

In Theorems 5 and 6 we will show that graphs for which the bounds of Theorem 2 are tight up to a constant factor exist. Note that for $g(i) = (i + 1)^\alpha$, where $\alpha > 1$, one has $A = 1 + 2 \sum_{i=2}^\infty \frac{1}{g(i-1)} < \infty$. For example, $A = 1 + \frac{\pi^2}{3}$ for $\alpha = 2$ and $A = 1 + \frac{\pi^4}{45}$ for $\alpha = 4$.

If G is a graph of maximum degree d , then $G \in LS(g, bw(G))$, where $g(i) = (d - 1)^i$, because $\max\{|V_0^i|, |V_1^i|\} \leq bw(G)(d - 1)^i$. Thus, Theorem 2 implies that

$$bw(G) \geq A \frac{\lambda_2 n}{4} (1 - o(1)) = \frac{d}{d-2} \frac{\lambda_2 n}{4} (1 - o(1)) \text{ as } \frac{n}{bw(G)} \rightarrow \infty. \quad (11)$$

It shows that the traditional spectral bound of Eq. (1) can be asymptotically improved by a factor of 3 for 3-regular graphs and by a factor of 2 for 4-regular graphs provided that $\frac{n}{bw}$ is very large. In Theorem 8, we show that this bound is asymptotically tight for double-root trees.

For the next, we derive bounds for sufficiently large 3- and 4-regular graphs. These bounds also hold for graphs where $bw = \Theta(n)$, i.e. when the number of levels does not necessarily increase with the number of vertices. One example are the d -regular Ramanujan graphs. For these graphs it holds that $\lambda_2 \geq d - 2\sqrt{d-1}$. Notice that the latter value is asymptotically the largest possible value for d -regular graphs, since $\lambda_2 \leq d - 2\sqrt{d-1} + \frac{4\sqrt{d-1}}{\log_{d-1}(n) - O(1)}$ [19] (see also [1,14]).

The classical spectral bound of Eq. (1) implies a lower bound for the bisection width which takes the form $0.042n$ for 3-regular Ramanujan graphs and $0.133n$ for 4-regular Ramanujan graphs. First we compute new lower bounds for d -regular graphs with a small λ_2 .

Theorem 3. *The bisection width of any*

1. *3-regular graph with $\lambda_2 \leq \frac{5-\sqrt{17}}{2}$ is at least $\min\{\frac{n}{6}, \frac{10+\lambda_2^2-7\lambda_2}{8+3\lambda_2^2-17\lambda_2^2+10\lambda_2} \frac{\lambda_2 n}{2}\}$.*
2. *d -regular graph with $\lambda_2 \leq 2$ is at least $\min\{\frac{n}{2}, \frac{d+1-\lambda_2}{2d-1-(\lambda_2-1)^2} \frac{\lambda_2 n}{2}\}$.*

Proof. As mentioned above, for any d -regular graph G it holds that $G \in LS(g, bw)$, where $g(i) = (d-1)^i$, for any l such that $n \geq 2bw \sum_{i=1}^{l-1} g(i-1)$. By solving the equation of Lemma 1 for bw we get

$$bw \geq \max_{1=a_1 \leq \dots \leq a_l} \frac{a_l^2 \lambda_2 \frac{n}{2}}{2a_1^2 + \sum_{i=1}^{l-1} [(d-1)^i (a_i - a_{i+1})^2 + \lambda_2 (d-1)^{i-1} (a_i^2 - a_{i+1}^2)]}. \quad (12)$$

In the sequel we use the values

$$a_1 = 1, \quad a_2 = \frac{1+d-\lambda_2}{d-1}, \quad a_{j+1} = a_j \frac{d-\lambda_2}{d-1} - a_{j-1} \frac{1}{d-1} \text{ for } j > 2$$

such that $1 = a_1 \leq a_2 \leq \dots \leq a_l$.

For the next we discuss each case separately.

3-regular, $\lambda_2 \leq \frac{5-\sqrt{17}}{2}$ and $bw \leq \frac{n}{6}$: We apply Eq. (12) using $d=3$ and $l=3$. We have $1 = a_1 \leq a_2 = \frac{4-\lambda_2}{2} \leq a_3 = \frac{\lambda_2^2-7\lambda_2+10}{4}$ and $n \geq 2bw(g(0) + g(1))$.

We get $bw \geq \frac{(\lambda_2^2-7\lambda_2+10)\lambda_2}{3\lambda_2^3-17\lambda_2^2+10\lambda_2+8} \frac{n}{2}$ in this case.

d -regular, $\lambda_2 \leq 2$ and $bw \leq \frac{n}{2}$: We apply Eq. (12) using $l=2$. We have $1 = a_1 \leq a_2 = \frac{1+d-\lambda_2}{d-1}$ and $n \geq 2bwg(0)$.

We get $bw \geq \frac{(d+1-\lambda_2)\lambda_2}{2(d-1)-\lambda_2^2+2\lambda_2} \frac{n}{2}$ in this case. $\square$

From $\lambda_2 \leq d - 2\sqrt{d-1} + \frac{4\sqrt{d-1}}{\log_{d-1}(n) - O(1)}$ [1,14,19] it follows that for $d=3$ it is $\lambda_2 \leq 0.17 + o(1)$, for $d=4$ it is $\lambda_2 \leq 0.54 + o(1)$ and for $d=5$ it is $\lambda_2 \leq 1 + o(1)$.

Thus, the conditions $\lambda_2 \leq \frac{5-\sqrt{17}}{2}$ and $\lambda_2 \leq 2$ of cases 1 and 2 of Theorem 3 hold for reasonable degrees and for sufficiently large graphs.

An upper bound of $(0.4 + \delta)n$ for the bisection width of sufficiently large 4-regular graphs and an upper bound of $(\frac{1}{6} + \delta)n$ for the bisection width of sufficiently large 3-regular graphs are shown in [18].

Now we compare the bounds of Theorem 3 with the traditional spectral bound $\frac{\lambda_2 n}{4}$ of Eq. (1). We observe that the factor $\frac{d+1-\lambda_2}{2d-1-(\lambda_2-1)^2}$ in front of $\frac{\lambda_2 n}{2}$ is always greater than $\frac{1}{2}$, so Theorem 3 provides us with better lower bounds on the bisection width of d -regular graphs than the classical spectral lower bound. This result is improved in the first case of Theorem 3 for 3-regular graphs by using 3 levels instead of 2 levels.

Theorem 3 can also be used to derive stronger lower bounds on the bisection width of Ramanujan graphs. The function $\frac{10+\lambda_2^2-7\lambda_2}{8+3\lambda_2^2-17\lambda_2+10\lambda_2}\lambda_2$ is monotone increasing in the interval $[0, \frac{5-\sqrt{17}}{2}]$ and the function $\frac{5-\lambda_2}{7-(\lambda_2-1)^2}\lambda_2$ is monotone increasing in the interval $[0, 2]$. Since $\lambda_2 \geq 3 - 2\sqrt{2} \approx 0.171573$ for 3-regular Ramanujan graphs and $\lambda_2 \geq 4 - 2\sqrt{3} \approx 0.535898$ for 4-regular Ramanujan graphs, Theorem 3 leads to the following corollary.

Corollary 1. *The bisection width of any sufficiently large 3-regular Ramanujan graph (such that $\lambda_2 \leq \frac{5-\sqrt{17}}{2}$) is at least $0.082n$. The bisection width of any sufficiently large 4-regular Ramanujan graph (such that $\lambda_2 \leq 2$) is at least $0.176n$.*

This is an improvement on the previously known lower bounds of $0.042n$ for 3-regular and of $0.133n$ for 4-regular Ramanujan graphs, which can be derived by using the traditional spectral bound.

What remains to be said is that the results of Theorem 3 cannot generally be improved by using more levels. Consider for example 4-regular graphs where we only used 2 levels. We cannot take $l \geq 3$, because when $l=3$, Lemma 1 only holds if $n > 2bw \sum_{i=1}^2 g(i-1) = 8bw$, which is not generally fulfilled in the case of 4-regular graphs (Ramanujan graphs are counterexamples). Using the same arguments, we cannot take $l \geq 4$ for 3-regular graphs. When $l=4$ Lemma 1 only holds if $n > 2bw \sum_{i=1}^3 g(i-1) = 14bw$, which is false for 3-regular Ramanujan graphs.

4. Worst case graphs

In this section we will present examples to show that the results of the previous section are asymptotically tight. First, we introduce a new class of edge-weighted graphs. They belong to the class $LS(g, \sigma)$ where $g(i) = (i+1)^\alpha$. We show that the results of Theorem 2 are tight for these graphs if $\alpha \geq 1$ and tight up to a constant factor if $0 \leq \alpha < 1$. Second, we analyze the double-root trees without edge weights as a more realistic example. Again, we can show that the results of the previous section are tight.

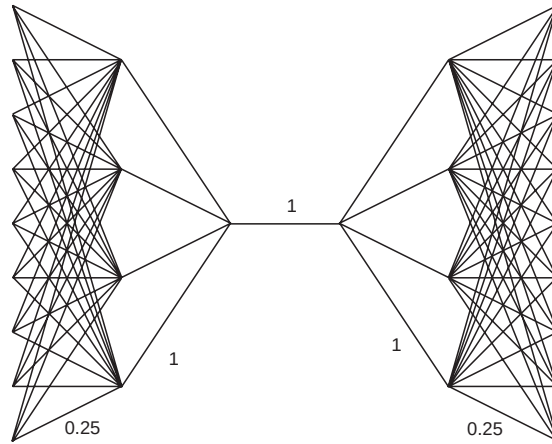


Fig. 2. The structure of the graph B_3^2 . The values indicate the weight of the edges.

4.1. Worst case graphs with edge weights

The Laplacian of a graph can be generalized to edge weighted graphs where the off-diagonal entries contain the negative values of the corresponding edge weight. The results of the previous sections and, particularly Theorem 2 can also be applied to edge weighted graphs.

Definition 2. The edge-weighted graph, which is denoted as B_l^α , $\alpha \geq 0$ and $l \geq 1$, is obtained as follows. B_l^α consists of two symmetric sides. Each side consists of l levels and level i contains i^α vertices, $1 \leq i \leq l$. Edges connect each pair of vertices in consecutive levels. Each edge between the vertices in levels i and $i + 1$ is weighted by $\frac{1}{i^\alpha}$. Moreover, the single vertices on level 1 of each side are connected to each other by an edge of weight 1.

The graphs B_l^α belong to the class $LS(g, \sigma)$, where $g(i) = (i + 1)^\alpha$. An example where $\alpha = 2$ and $l = 3$ is presented in Fig. 2. Throughout this section, $\lambda_2(B_l^\alpha)$ denotes the second smallest eigenvalue of the Laplacian of B_l^α . In order to compute $\lambda_2(B_l^\alpha)$, we need the following lemmas.

Lemma 2. A matrix with the structure $\begin{pmatrix} M_1 & M_2 \\ M_2 & M_1 \end{pmatrix}$ has the same eigenvalues as the matrix $\begin{pmatrix} M_1 + M_2 & 0 \\ 0 & M_1 - M_2 \end{pmatrix}$.

Proof. Let x be an eigenvector of the matrix $M_1 + M_2$. Then (x, x) will be an eigenvector of the matrix $\begin{pmatrix} M_1 & M_2 \\ M_2 & M_1 \end{pmatrix}$ with the same eigenvalue. If y is an eigenvector of the matrix $M_1 - M_2$ then $(y, -y)$ will be an eigenvector of the matrix $\begin{pmatrix} M_1 & M_2 \\ M_2 & M_1 \end{pmatrix}$ with the same eigenvalue. $\square$

In order to prove Lemma 3 we need the so called Separation Theorem (see [22]).

Theorem 4. For a real symmetric matrix A_n it holds that the eigenvalues $\lambda'_1, \dots, \lambda'_{n-1}$ of any principal minor A_{n-1} of the matrix A_n separate the eigenvalues $\lambda_1, \dots, \lambda_n$ of A_n .

We can now prove the following lemma.

Lemma 3. If $\lambda_2(B_l^z) < 2$, then the entries in the eigenvector of B_l^z with the eigenvalue λ_2 are equal for all vertices on the same level. Moreover $x_i^0 = -x_i^1$, with x_i^s being the entry of the corresponding eigenvector on level i on side $s \in \{0, 1\}$.

Proof. We consider two vertices $i1$ and $i2$ on level i on either side. Let x_{i1} and x_{i2} be their entries in the eigenvector of B_l^z with an eigenvalue λ where $x_{i1} \neq x_{i2}$. Both vertices are adjacent to the same vertices on levels $i-1$ and $i+1$. The entry of the eigenvector which corresponds to a vertex q on level i , $1 \leq q \leq i^z$ is denoted as x_{iq} . We obtain

$$-\sum_{q=1}^{(i-1)^z} x_{(i-1)q} + \deg_i x_{i\varepsilon} - \sum_{q=1}^{(i+1)^z} \frac{g(i)}{g(i-1)} x_{(i+1)q} = \lambda x_{i\varepsilon},$$

i.e.

$$(\deg_i - \lambda)x_{i\varepsilon} = \sum_{q=1}^{(i-1)^z} x_{(i-1)q} + \sum_{q=1}^{(i+1)^z} \frac{g(i)}{g(i-1)} x_{(i+1)q}$$

where $\varepsilon \in \{1, 2\}$ and $\deg_i$ is the degree of a vertex on level i . Then $x_{i1} \neq x_{i2}$ implies $\lambda = \deg_i = 1 + (\frac{i+1}{i})^z > \lambda_2$. Therefore, if $x_{i1} \neq x_{i2}$ then the corresponding eigenvalue is not the second smallest of our Laplacian.

We arrange the vertices of the graph in the Laplacian in the following way. We set the vertices on the left side of the bisection in the first $n/2$ rows where we begin with the vertex which is incident to the bisection edge. The next $n/2$ rows represent the vertices on the right hand side of the bisection, and we use the same order for them as we did for the first $n/2$ vertices. Then the Laplacian of B_l^z has the structure $\begin{pmatrix} M_1 & M_2 \\ M_2 & M_1 \end{pmatrix}$, and its eigenvalues are the union of the eigenvalues of the matrices $M_1 + M_2$ and $M_1 - M_2$ (see Lemma 2). The first entry of M_2 is a -1 and all other entries are 0.

The matrices which are obtained from $M_1 + M_2$ and from $M_1 - M_2$ by deleting the first row and the first column are denoted as $(M_1 + M_2)(1, 1)$ and $(M_1 - M_2)(1, 1)$, respectively. Using the Separation Theorem it is clearly evident that the eigenvalues of $(M_1 + M_2)(1, 1)$ separate the eigenvalues of $M_1 + M_2$ and that the eigenvalues of $(M_1 - M_2)(1, 1)$ separate the eigenvalues of $M_1 - M_2$. However $(M_1 + M_2)(1, 1) = (M_1 - M_2)(1, 1)$ and 0 is an eigenvalue of $(M_1 + M_2)$. Hence, the smallest eigenvalue of $M_1 - M_2$ is smaller than the second smallest eigenvalue of $M_1 + M_2$. Therefore, since the smallest eigenvalue of $M_1 + M_2$ is 0 (which is the smallest eigenvalue of the original matrix) and this eigenvalue is simple, then the second smallest eigenvalue of

the Laplacian of B_l^z is the smallest eigenvalue of $M_1 - M_2$ and, as shown in Lemma 2, $x_i^1 = -x_i^0$ holds for all $1 \leq i \leq l$. $\square$

As described in Lemma 3, any entry on the same level of the eigenvector of B_l^z with the second smallest eigenvalue has the same value and moreover $x_i^0 = -x_i^1$ for all $i \in \mathbb{N}$. Therefore, $\lambda_2(B_l^z)$ can be obtained from the eigenvalues of a certain weighted path.

Now we are going to calculate lower bounds for $\lambda_2(B_l^z)$, and it follows from Theorem 2 that an l_0 exists, such that it is $\lambda_2(B_l^z) < 2$ for any $l \geq l_0$. For the next we consider $l \geq l_0$. We denote the entry of the eigenvector with $\lambda_2(B_l^z)$ on level i as x_i . Now we set $x_i = a_i + r_i$, where $r_i = -\lambda_2(B_l^z)p_i$ and $a_i = 1 + 2 \sum_{j=1}^{i-1} \frac{1}{g(j)}$. Here r_i represents an error and our method consists of proving that the r_i 's are small compared to a_i .

From the equality $(g(i-1) + g(i))x_i - g(i-1)x_{i-1} - g(i)x_{i+1} = \lambda_2(B_l^z)g(i-1)x_i$, $2 \leq i \leq l-1$ and from $(g(1) + 1)x_1 + x_1 - g(1)x_2 = \lambda_2(B_l^z)x_1$ it follows that

$$p_{i+1} = p_i + \alpha(i)((1 - \lambda_2)p_i - p_{i-1} + a_i), \quad (13)$$

where $p_1 = 0$, $p_2 = \frac{1}{g(1)}$ and $\alpha(i) = \frac{g(i-1)}{g(i)}$.

First, we consider the case $\alpha > 1$ and let $A = 1 + 2 \sum_{j=1}^{\infty} \frac{1}{g(j)} < \infty$. Let q_i be defined as $q_{i+1} = q_i + \alpha(i)(q_i - q_{i-1} + A)$, $q_1 = 0$, $q_2 = \frac{1}{g(1)}$. From the definition of q_i , we obtain $q_i = \sum_{j=1}^{i-1} \frac{1}{g(j)} + \sum_{(p,q), 1 \leq p < q \leq i-1} \frac{g(p)}{g(q)} A$ and $q_{i+1} - q_i = \frac{1}{g(i)} + \sum_{p=1}^{i-1} \frac{g(p)}{g(i)} A$.

Lemma 4. For the graph B_l^z where $\alpha > 1$ it holds that $q_{i+1} - q_i \geq p_{i+1} - p_i \geq 0$ for any $1 \leq i \leq l-1$. Moreover, $\lambda_2(B_l^z)q_l \rightarrow 0$

Proof. First we show the second statement of our lemma. It holds that

$$\begin{aligned} q_i &= \sum_{j=1}^{i-1} \frac{1}{g(j)} + \sum_{1 \leq p < q \leq i-1} \frac{g(p)}{g(q)} A \leq A + \sum_{q=1}^{i-1} \sum_{p=1}^{q-1} \frac{g(p)}{g(q)} A \\ &\leq A + \sum_{p=1}^{i-1} \left(\sum_{q=p+1}^{i-1} \frac{1}{g(q)} \right) g(p)A. \end{aligned}$$

Using the result of Theorem 2, it holds that $\lambda_2(B_l^z) \leq O(l^{-2})$ and therefore the second statement of the lemma holds. For the first statement let $\delta_i = q_i - p_i$. We use induction on i . For $i = 1$ the lemma holds. We assume that the lemma holds for any $j \leq i$. Then

$$\begin{aligned} \delta_{i+1} &= q_{i+1} - p_{i+1} = q_i + \alpha(i)(q_i - q_{i-1} + A) - p_i \\ &\quad - \alpha(i)((1 - \lambda_2(B_l^z))p_i - p_{i-1} + a_i) \\ &= \delta_i + \alpha(i)(\delta_i - \delta_{i-1}) + \alpha(i)(A - a_i) + \alpha(i)\lambda_2(B_l^z)p_i \geq \delta_i. \end{aligned}$$

Now we have to show that $p_{i+1} - p_i \geq 0$. Our assumption was that the lemma holds for any j , $0 \leq j \leq i$. This implies that $0 \leq p_i \leq q_i$. Then, Eq. (13) implies that $p_{i+1} - p_i \geq \alpha(i)(a_i - \lambda_2(B_l^z)q_i) \geq 0$, where $\lambda_2(B_l^z) \leq O(l^{-2})$ due to Theorem 2. Thus, the last inequality also holds. $\square$

In the next theorem we state a lower bound for $\lambda_2(B_l^\alpha)$, where $A = 1 + 2 \sum_{j=1}^{\infty} \frac{1}{g(j)} < \infty$.

Theorem 5. *If $\alpha > 1$ then it holds that*

$$\lambda_2(B_l^\alpha) \geq \frac{4}{A} \frac{1}{n(1 + o(1))}.$$

Proof. Let us consider $\lambda_2(B_l^\alpha) = \frac{f_1(x)}{f_2(x)}$, where x is the eigenvector to the eigenvalue $\lambda_2(B_l^\alpha)$. Now, Lemma 3 implies that $x_i^0 = -x_i^1$ and therefore it holds that $f_1(x) = 4x_1^2 + 2 \sum_{i=1}^{l-1} (x_{i+1} - x_i)^2 g(i)$ and $f_2(x) = 2 \sum_{i=1}^l x_i^2 g(i-1)$. We get

$$\begin{aligned} f_1(x) &= 4 + 2 \sum_{i=1}^{l-1} \left(2 \frac{1}{g(i)} - \lambda_2(B_l^\alpha)(p_{i+1} - p_i) \right)^2 g(i) \\ &\geq 4 + 2 \sum_{i=1}^{l-1} \frac{4}{g(i)} - 8\lambda_2(B_l^\alpha) \sum_{i=1}^{l-1} (q_{i+1} - q_i) \geq 4a_l - o(l) \end{aligned}$$

and

$$\begin{aligned} f_2(x) &\leq 2 \sum_{i=1}^{l-1} (a_i^2 + \lambda_2^2(B_l^\alpha) q_i^2) g(i) \leq 2 \sum_{i=1}^{l-1} a_i^2 g(i) + o(l^2) \\ &\leq 2 \sum_{i=1}^{l-1} a_i^2 g(i) + o(l^2). \end{aligned}$$

By calculating the bounds for $f_1(x)$ and $f_2(x)$ we obtain the result of the theorem. $\square$

This theorem can be easily generalized to any function g , such that $A < \infty$.

We are now ready to discuss the case $0 \leq \alpha < 1$. We define q_i as $q_{i+1} = q_i + \alpha(i)(q_i - q_{i-1} + a_i)$, where $q_1 = 0$ and $q_2 = \frac{1}{g(1)}$. We obtain $q_i = \sum_{j=1}^{i-1} \frac{1}{g(j)} + \sum_{p=1}^{i-2} \sum_{q=p+1}^{i-1} \frac{g(p)}{g(q)} a_{p+1}$.

Lemma 5. *For $0 \leq \alpha < 1$ we obtain $q_i \leq \frac{1}{(1-\alpha)(3-\alpha)} i^{3-\alpha} (1 + O(i^{\alpha-1}))$. Moreover $q_{i+1} - q_i \geq p_{i+1} - p_i \geq 0$ for a sufficiently large l , where $1 \leq i \leq l-1$.*

Proof. First we set $\sigma_i = \sum_{j=1}^{i-1} \frac{1}{(j+1)^\alpha} \leq \frac{1}{1-\alpha} (i+1)^{1-\alpha}$. So we get

$$q_i = O(i^{1-\alpha}) + \sigma_i \sum_{p=1}^{i-2} g(p) a_{p+1} - \sum_{p=1}^{i-2} \sigma_{p+1} g(p) a_{p+1} = \frac{1}{(1-\alpha)(3-\alpha)} i^{3-\alpha}.$$

For the second statement of the lemma we use induction on i . For $i=1$ the lemma holds. Now we assume that the lemma is true for any $j \leq i$. Then $q_{i+1} - q_i \geq p_{i+1} - p_i$

is equivalent to $q_{i+1} - p_{i+1} \geq q_i - p_i$. We get

$$\begin{aligned} q_{i+1} - p_{i+1} &= q_i + \alpha(i)(q_i - q_{i-1} + a_i) - p_i - \alpha(i)((1 - \lambda_2(B_l^\alpha))p_i - p_{i-1} + a_i) \\ &= q_i - p_i + \alpha(i)((q_i - p_i) - (q_{i-1} - p_{i-1}) + \lambda_2(B_l^\alpha)p_i) \geq q_i - p_i. \end{aligned}$$

We assume that the lemma holds for $1 \leq j \leq i$ and therefore $0 \leq p_i \leq q_i$. Then we obtain

$$\begin{aligned} p_{i+1} - p_i &= \alpha(i)(p_i - p_{i-1} + a_i - \lambda_2(B_l^\alpha)p_i) = \alpha(i)(p_i - p_{i-1}) \\ &\quad + \alpha(i)(a_i - \lambda_2(B_l^\alpha)p_i). \end{aligned}$$

We have to show that $a_i - \lambda_2(B_l^\alpha)p_i \geq 0$. Using Theorem 2 and the first statement of the lemma we obtain

$$a_i - \lambda_2(B_l^\alpha)p_i \geq \left(1 - \frac{1 - \alpha}{2 - \alpha} \frac{i^2(1 + O(i^{\alpha-1}))}{n^2(1 + o(1))}\right) a_i \geq 0. \quad \square$$

Now we are ready to calculate a lower bound for $\lambda_2(B_l^\alpha)$.

Theorem 6. If $0 \leq \alpha < 1$ then a constant $c(\alpha)$ exists such that $\lambda_2(B_l^\alpha) \geq c(\alpha) \frac{1}{l^2(1+o(1))}$ where $c(\alpha) = \frac{2(1-\alpha^2)(3-\alpha)(7-\alpha)}{4(7-\alpha)+(1-\alpha)^2(3-\alpha)}$.

Proof. Let $\lambda_2(B_l^\alpha) = \frac{f_1(x)}{f_2(x)}$, where $f_1(x) \geq 4a_l - 4\lambda_2(B_l^\alpha) \sum_{i=1}^{l-1} (q_{i+1} - q_i)$ and $f_2(x) \leq 2 \sum_{i=1}^{l-1} (a_i^2 + \lambda_2^2(B_l^\alpha) q_i^2) g(i)$. We omit the low-order terms and we obtain

$$\begin{aligned} q_i - q_{i-1} &= \frac{1}{g(i-1)} + \sum_{p=1}^{i-2} \frac{g(p)}{g(i-1)} a_{p+1} \\ &= O(i^{-\alpha}) + \frac{1}{i^\alpha} \sum_{p=1}^{i-2} (p+1)^\alpha \frac{2}{1-\alpha} (p+1)^{1-\alpha} \\ &= O(i^{-\alpha}) + \frac{1}{i^\alpha} \frac{2}{1-\alpha} \frac{i^2}{2} = \frac{1}{1-\alpha} i^{2-\alpha} + O(i^{-\alpha}) \end{aligned}$$

and by using the upper bound of Theorem 2 on $\lambda_2(B_l^\alpha)$ we get

$$\lambda_2(B_l^\alpha) \sum_{i=1}^{l-1} (q_{i+1} - q_i) \leq \frac{(1-\alpha)(3-\alpha)}{l^2} \frac{1}{(1-\alpha)(3-\alpha)} l^{(3-\alpha)}.$$

Then we get

$$f_1(x) \geq 4 \left(\frac{2}{1-\alpha} l^{1-\alpha} - l^{1-\alpha} \right) = 4 \frac{1+\alpha}{1-\alpha} l^{1-\alpha}.$$

On the other side, when we use Lemma 5, we obtain

$$\begin{aligned} f_2(x) &\leq 2 \sum_{i=1}^l a_i^2 g(i) + 2\lambda_2^2(B_l^\alpha) \sum_{i=1}^l q_i^2 g(i) \\ &= 2 \left(\frac{4}{(1-\alpha)^2(3-\alpha)} l^{3-\alpha} + \frac{1}{l^4(7-\alpha)} l^{7-\alpha} \right). \quad \square \end{aligned}$$

For the next we consider the case $\alpha = 1$ and we define q_i as $q_{i+1} = q_i + \alpha(i)(q_i - q_{i-1} + a_i)$, where $q_1 = 0$ and $q_2 = \frac{1}{g(1)}$. We obtain $q_i = \sum_{j=1}^{i-1} \frac{1}{g(j)} + \sum_{p=1}^{i-2} \sum_{q=p+1}^{i-1} \frac{g(p)}{g(q)} a_{p+1}$.

Lemma 6. *Let $g(i) = (i+1)$ and $\alpha = 1$. Then $q_{i+1} - q_i \geq p_{i+1} - p_i \geq 0$ for any $1 \leq i \leq l-1$. Moreover $0 \leq q_i \leq (1 + \frac{1}{4} i^2) a_i$ for any $1 \leq i \leq l-1$.*

Proof. First we show the second statement of our lemma:

$$\begin{aligned} q_i &= \sum_{j=1}^{i-1} \frac{1}{g(j)} + \sum_{1 \leq p < q \leq i-1} \frac{g(p)}{g(q)} a_i \leq a_i + \sum_{q=1}^{i-1} \sum_{p=1}^{q-1} \frac{g(p)}{g(q)} a_i \\ &\leq a_i + \sum_{q=1}^{i-1} \frac{q}{2} a_i \leq a_i + \frac{i^2}{4} a_i. \end{aligned}$$

For the first statement let $\delta_i = q_i - p_i$. We use induction on i . For $i = 1$ the lemma holds. We assume that the lemma is true for any $j \leq i$. Then

$$\begin{aligned} \delta_{i+1} &= q_{i+1} - p_{i+1} = q_i + \alpha(i)(q_i - q_{i-1} + a_i) - p_i \\ &\quad - \alpha(i)((1 - \lambda_2(B_l^\alpha))p_i - p_{i-1} + a_i) \\ &= \delta_i + \alpha(i)(\delta_i - \delta_{i-1}) + \alpha(i)\lambda_2(B_l^\alpha)p_i \geq \delta_i. \end{aligned}$$

Now we have to show that $p_{i+1} - p_i \geq 0$. Our assumption was that the lemma holds for any j , $0 \leq j \leq i$. This implies that $0 \leq p_i \leq q_i$. Then $p_{i+1} - p_i \geq \alpha(i)(a_i - \lambda_2(B_l^\alpha)q_i) \geq \alpha(i)(a_i - \lambda_2(B_l^\alpha)a_i - \lambda_2(B_l^\alpha)\frac{i^2}{4}a_i) \geq 0$, where $\lambda_2(B_l^\alpha) = O\left(\frac{1}{l^2 \ln(l)}\right)$ because of Theorem 2 and therefore the last inequality is also fulfilled. $\square$

In the next theorem we state a lower bound for $\lambda_2(B_l^1)$.

Theorem 7. *It holds that*

$$\lambda_2(B_l^1) \geq \frac{2}{l^2 \ln(l)(1 + o(1))}.$$

Proof. Let us consider $\lambda_2(B_l^1) = \frac{f_1(x)}{f_2(x)}$, where x is the eigenvector to the eigenvalue $\lambda_2(B_l^1)$. Using Lemma 3, $f_1(x) = 4x_1^2 + 2 \sum_{i=1}^{l-1} (x_{i+1} - x_i)^2 g(i)$ and $f_2(x) = 2 \sum_{i=1}^l x_i^2 g(i-1)$. We get

$$\begin{aligned} f_1(x) &= 4 + 2 \sum_{i=1}^{l-1} \left(2 \frac{1}{g(i)} - \lambda_2(B_l^1)(p_{i+1} - p_i) \right)^2 g(i) \\ &\geq 4 + 2 \sum_{i=1}^{l-1} \frac{4}{g(i)} - 8\lambda_2(B_l^1) \sum_{i=1}^{l-1} (q_{i+1} - q_i) \geq 4 \ln(l)(1 - o(1)) \end{aligned}$$

and

$$\begin{aligned} f_2(x) &\leq 2 \sum_{i=1}^{l-1} (a_i^2 + \lambda_2^2(B_l^1)q_i^2)g(i) \leq 2 \sum_{i=1}^{l-1} a_i^2 g(i) + o(1) \\ &\leq 2l^2 \ln^2(l)(1 + o(1)). \end{aligned}$$

By calculating the bounds for $f_1(x)$ and $f_2(x)$, we obtain the result of the theorem. $\square$

We will summarize the results of this section and compare them with the results of Theorem 2. For the case $\alpha > 1$, the upper and lower bounds on λ_2 of Theorems 2 and 5 are asymptotically tight. For the case $\alpha = 1$, the bounds are also asymptotically tight, since the lower bound on λ_2 of Theorem 7, depending on l , is the same as the upper bound shown in the proof of Theorem 2. In the case $0 < \alpha < 1$, the bounds of Theorems 2 and 6 are tight up to a constant factor, since it is known that $n = 2 \sum_{i=1}^l g(i) = \frac{2}{\alpha+1} l^{\alpha+1}(1 + o(1))$ and thus a $d(\alpha)$ exists such that $\lambda_2 \geq d(\alpha) \frac{1}{n^{\frac{\alpha}{\alpha+1}(1+o(1))}}$. A special case occurs when $\alpha = 0$. The graph B_l^0 is a path of length $2l$ and we get $c(0) = \frac{42}{31}$ from Theorem 6. From Theorem 2, we obtain $\lambda_2 \leq 3 \frac{1}{l^2}(1 + o(1))$. However, in the case of the path we know that $\lambda_2 \approx \frac{\pi^2}{4l^2}$. Thus, a gap of a constant factor exists between the lower and the upper bound for λ_2 .

4.2. The double-root trees as worst case graphs

We define the double-root tree $T_{d,l}$ as follows. It has a central edge c , each vertex of the tree either has degree 1 or d , and the distance between c and any vertex of degree 1 is $l - 1$. Let $n = 2 \frac{(d-1)^l - 1}{d-2}$ denote the number of vertices of $T_{d,l}$ and we denote the Laplacian of $T_{d,l}$ as $L(T_{d,l})$.

In this paper we make use of the proof of a lemma in [11], where a lower and upper bound is given for the second smallest eigenvalue of the double-root tree. In [11] the authors show that the entries of the eigenvector to the second smallest eigenvalue of the Laplacian of $T_{d,l}$ only depend on the number of levels l . Indeed for any $1 \leq i \leq l$ it holds that $x_i^0 = -x_i^1$. Moreover $T_{d,l}$ belongs to the class $LS(g, 1)$ where $g(i) = (d-1)^i$ and $A = 1 + 2 \sum_{j=1}^{i-1} \frac{1}{g(j)} < \infty$, and therefore we can apply the proof of Theorem 5 in order to obtain the following theorem

Theorem 8. *The smallest non-zero eigenvalue λ_2 of the Laplacian $L(T_{d,l})$ satisfies*

$$\lambda_2 = 4 \frac{d-2}{dn(1+o(1))} \quad \text{as } n \rightarrow \infty.$$

Theorem 8 shows that the bound of Eq. (11), which is derived from Theorem 2, is asymptotically tight.

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References

- [1] N. Alon, Eigenvalues and expanders, *Combinatorica* 6 (2) (1986) 83–96.
- [2] S.L. Bezroukov, R. Elsässer, B. Monien, R. Preis, J.-P. Tillich, New spectral lower bounds on the bisection width of graphs, in: *Workshop on Graph-Theoretic Concepts in Computer Science (WG)*, Lecture Notes in Computer Science, Vol. 1928, 2000, pp. 23–34.
- [3] B. Bollobas, The isoperimetric number of random regular graphs, *European J. Combin.* 9 (1988) 241–244.
- [4] T.N. Bui, S. Chaudhuri, F.T. Leighton, M. Sisper, Graph bisection algorithms with good average case behaviour, *Combinatorica* 7 (2) (1987) 171–191.
- [5] J. Cheeger, A lower bound for the smallest eigenvalue of the laplacian, *Problems Anal.* (1970) 195–199.
- [6] P. Chiu, Cubic ramanujan graphs, *Combinatorica* 12 (3) (1992) 275–285.
- [7] J. Dodziuk, W.S. Kendall, Combinatorial laplacians and isoperimetric inequality, *Pitman Res. Notes Math. Ser.* (1986) 68–74.
- [8] R. Elsässer, T. Lücking, B. Monien, On spectral bounds for the k -partitioning of graphs, *Theory Comput. Systems* 36 (2003) 461–478.
- [9] M. Fiedler, A property of eigenvectors of nonnegative symmetric matrices and its application to graph theory, *Czechoslovak Math. J. Praha* 25 (100) (1975) 619–633.
- [10] M.R. Garey, D.S. Johnson, L. Stockmeyer, Some simplified NP-complete graph problems, *Theoret. Comput. Sci.* 1 (1976) 237–267.
- [11] S. Guattery, G. Miller, On the performance of spectral graph partitioning methods, in: *Proc. 16th Annu. ACM-SIAM Symp. on Discrete Algorithms*, ACM-SIAM, 1995, pp. 233–242.
- [12] A.V. Kostochka, L.S. Melnikov, On a lower bound for the isoperimetric number of cubic graphs, in: V.F. Kolchin, et al. (Eds.), *Probabilistic Methods in Discrete Mathematics*, Proc. 3rd Internat. Petrozavodsk Conf., Progress in Pure and Applied Discrete Mathematics, Vol. 1, TVP/VSP, 1993, pp. 251–265.
- [13] F.T. Leighton, *Introduction to Parallel Algorithms and Architectures: Arrays, Trees, Hypercubes*, Morgan Kaufmann Publishers, Los Altos, CA, 1992.
- [14] A. Lubotzky, R. Phillips, P. Sarnak, Ramanujan graphs, *Combinatorica* 8 (3) (1988) 261–277.
- [15] G.A. Margulis, Explicit group-theoretical constructions of combinatorial schemes and their application to the design of expanders and concentrators, *Problems Inform. Transmission* 24 (1) (1988) 39–46.
- [16] B. Mohar, Isoperimetric numbers of graphs, *J. Combin. Theory* 47 (3) (1989) 274–291.
- [17] M. Morgenstern, Existence and explicit constructions of $q+1$ regular ramanujan graphs for every prime power q , *J. Combin. Theory, Ser. B* 62 (1) (1994) 44–62.
- [18] B. Monien, R. Preis, Bisection width of 3- and 4-regular graphs, in: *26th International Symposium on Mathematical Foundations of Computer Science (MFCS)*, Lecture Notes in Computer Science, Vol. 2136, 2001, pp. 524–536.
- [19] A. Nilli, On the second eigenvalue of a graph, *Discrete Math.* 91 (1991) 207–210.
- [20] A. Pothen, H.D. Simon, K.P. Liu, Partitioning sparse matrices with eigenvectors of graphs, *SIAM J. Matrix Anal. Appl.* 11 (3) (1990) 430–452.
- [21] D.A. Spielman, S.-H. Teng, Spectral partitioning works: planar graphs and finite element meshes, in: *Proc. 37th Conf. on Foundations of Computer Science*, 1996, pp. 96–105.
- [22] J.H. Wilkinson, *The Algebraic Eigenvalue Problem*, Oxford University Press, Oxford, 1965.